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Instrument Systems
KONICA MINOLTA Group

Measurement of Near Eye Display Using Goniophotometer and Imaging Light Measuring Device

Tianxing Zhu | 2025

Your Application – our Know-how!

Display Measurement

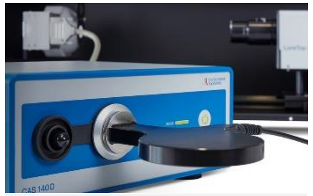

AR/VR


Automotive Interior Lighting

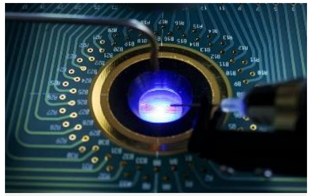

Display Characterization

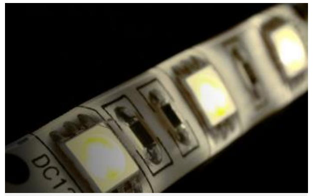

Flat Panel Display Production


MicroLEDs


System Audit

Light Emitter Measurement


LED Production


LED / Modules


UV-LEDs


VCSEL / Laser


IR Sensing

Lighting Products Measurement

Automotive Exterior Lighting


Blue Light Hazard

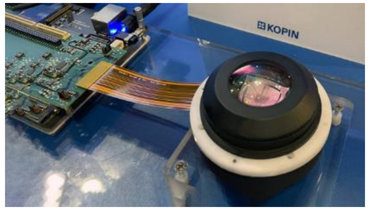

Solid State Lighting

AR/VR Testing Components

Final Headset

Alignment

Display-Lens Module & Optics

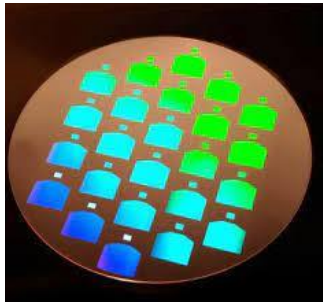


Microdisplay

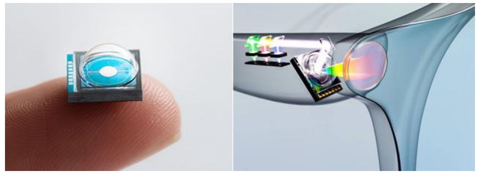
Final Headset

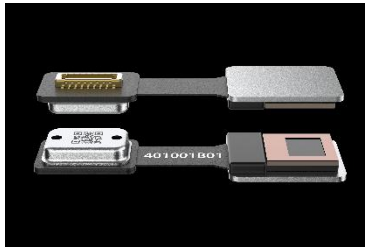
Prescription

Waveguide



Microdisplay / Projector





Agenda

1. Introduction
2. AR/VR Near-Eye-Display Measurement Systems
3. Measurement Setup
4. Result Comparison
5. Conclusions

Introduction

The quality/characteristic of a near eye display (NED) is important for the final performance of the VR/AR headset.

In this presentation two different measurement setups for near eye displays are compared:

- Goniophotometer DMS 803 (Instrument Systems)
- Imaging light measurement device (ILMD) LumiTop 5300 ARVR (Instrument Systems).

AR/VR Near-Eye-Display Measurement

CAS spectrometer as reference

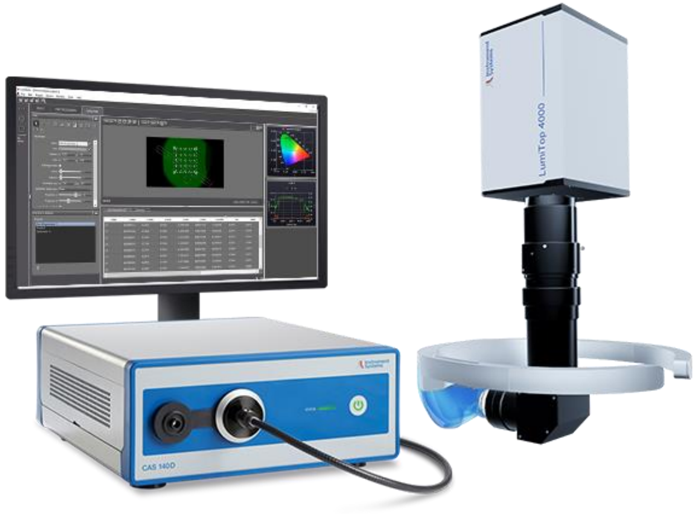
- Highest Spectral resolution
- Absolute calibration of luminance and color
- Traceable accuracy Y = ±3%, x/y = 0.0015

DMS Goniophotometer

- ✓ **Angle resolved spectral spot measurement**
- ✓ **Eyebox / Qualified Viewing space**
- ✓ **Time resolved**

TOP 300 AR/VR Optical Probe

- ✓ **Robust, compact and lightweight design for manufacturing applications**
- ✓ **High-accuracy luminance and color measurement**
- ✓ **Integrated viewfinder camera with 5 MP**
- ✓ **Optics mimic the human eye with a FOV of ±1.2°**
- ✓ **Binocular setup possible**



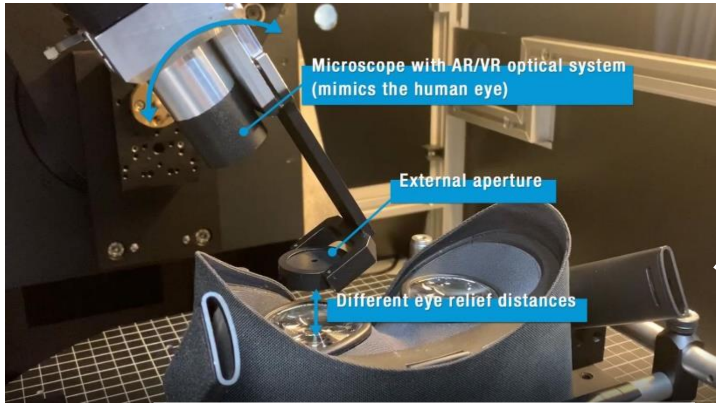
LumiTop AR/VR Imaging Light Measurement Device

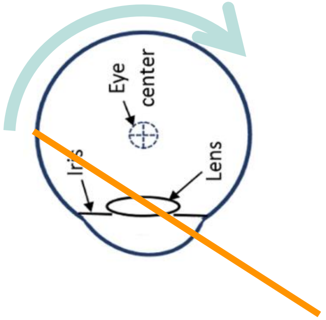
- ✓ **2D one-shot measurement at 120° FOV**
- ✓ **Distortion, Sharpness, Uniformity,Contrast, …**
- ✓ **Time resolved**
- ✓ **LumiSuite for integration and analysis**

DMS Goniophotometer – Reference System

DMS-803: Reference Goniophotometer

1. 6 motorized axes for automated measurements
   - Fully automated spectral sampling
   - Dedicated optics to mimic human visual system captures both: eye rotation and retinal image
   - Variety of accessories for ambient, specular and backlights are suitable for your angular measurements.
2. CAS 140D reference spectroradiometer
   - Spectral spot information for accurate measurement values
   - High-end CCD detector cooled and back illuminated

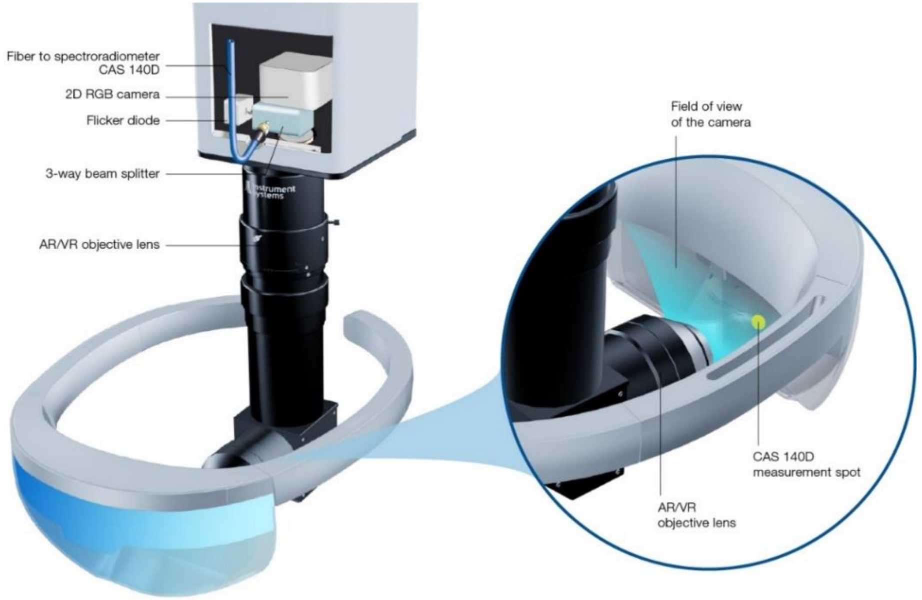




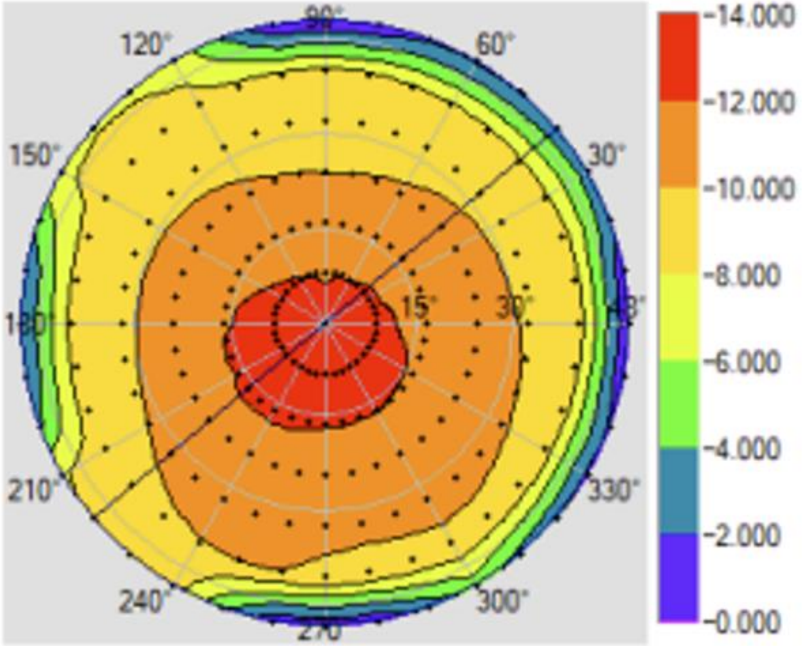
LumiTop AR/VR

LumiTop: 3-way beamsplitter (→ one-shot)

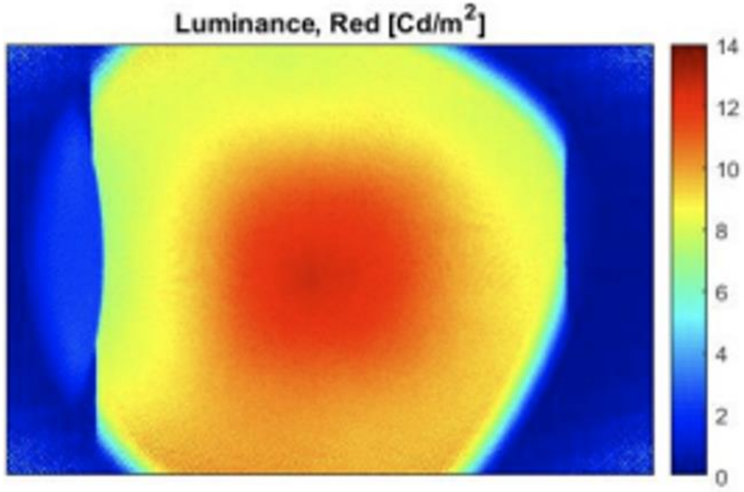
1. Camera system for 2D measurement
   - Calibrated System
   - 120° FOV & 40 px/deg.
2. CAS 140D reference spectroradiometer
   - Spectral spot information for live color fine-correction
   - High-end CCD detector cooled and back illuminated
3. Flicker diode
   - Sync to display modulation
   - Enables High Luminance mode



Differences between Spot and 2D Measurements



Luminance as seen by DMS angular resolved goniometric spot measurements.



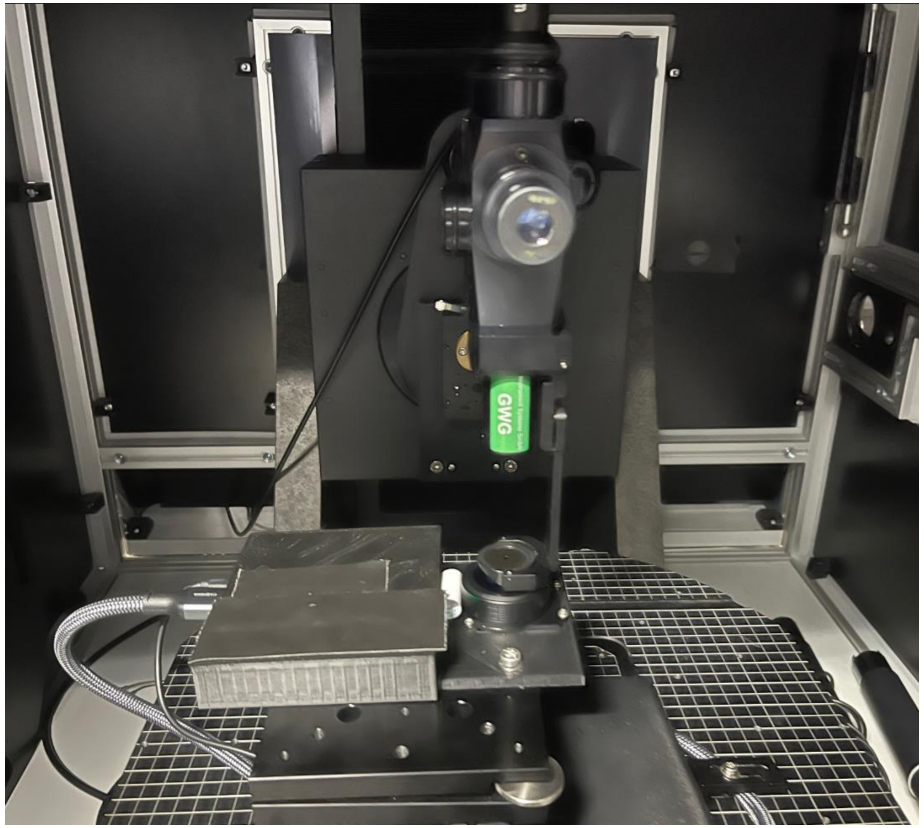
Luminance as seen by LumiTOP spectrally enhanced imaging colorimeter.

Measurements

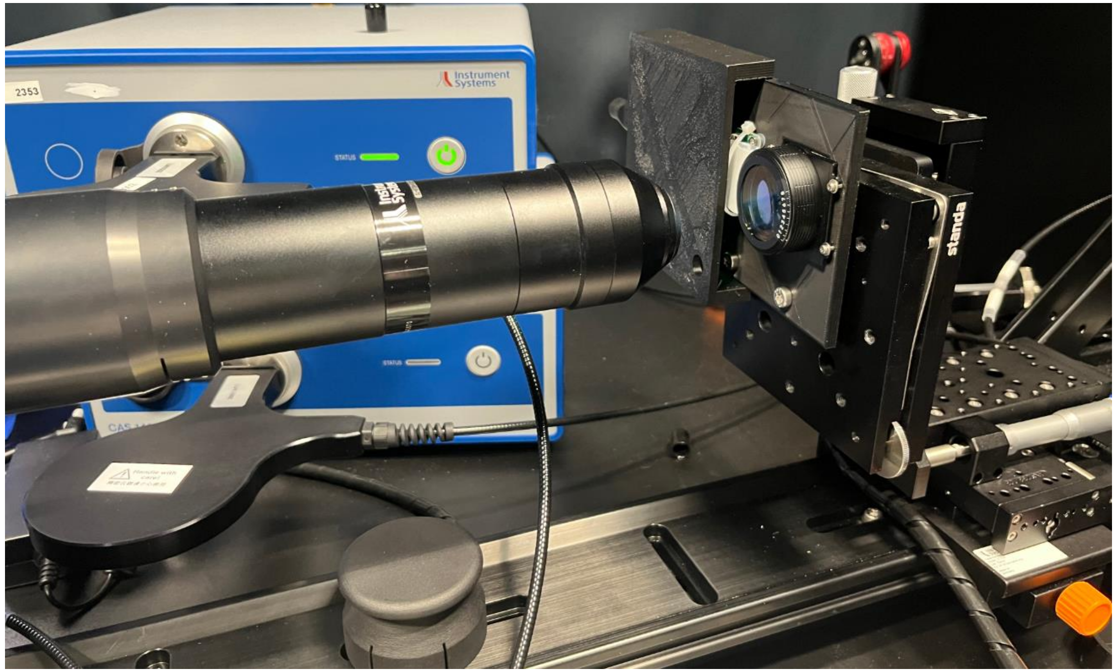
- Several characteristic are measured to obtain a perfect VR device
  - Final device
    - Luminance
    - Chromaticity
    - Gamma
    - Contrast
  - Display
    - Defects
    - Mura
    - Luminance
    - Chromaticity
    - Gamma
    - Contrast
  - Optical module (display+optics)
    - Distortion
    - MTF
    - FOV
    - VID
    - Chromaticity
    - Black/White
    - Grey 255
    - Contrast
    - Uniformity (Luminance/Color)
    - Chromatic aberration
    - Eye box
    - Veiling glare index

Measurement Setup

- DMS-803 Setup



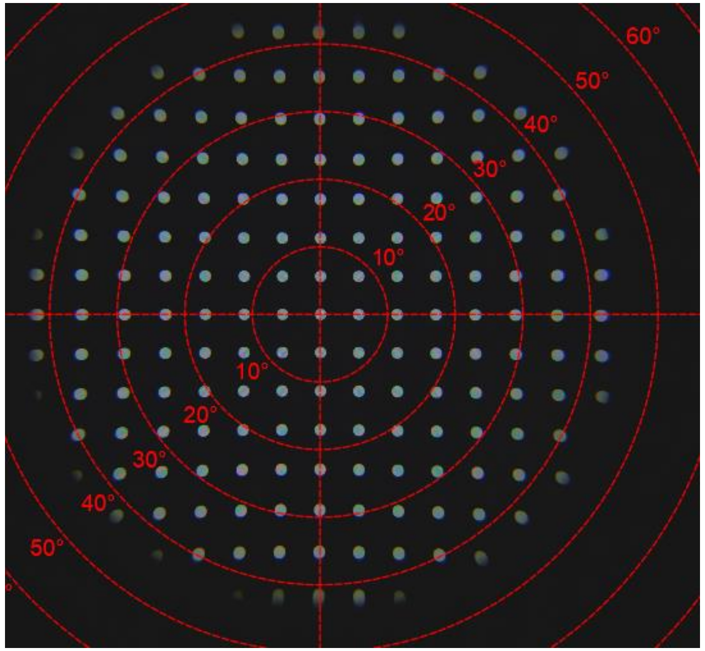
- LumiTop 5300 AR/VR

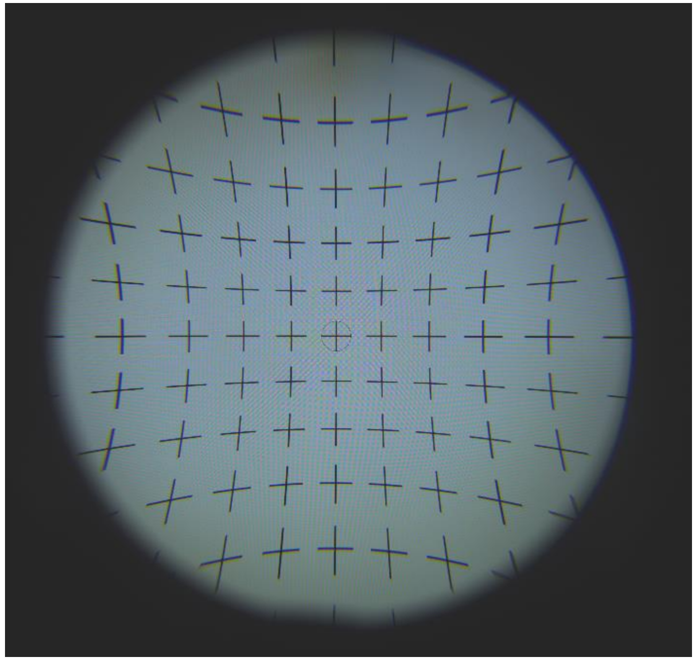


DUT: Pancake + LCD; 90° FOV

Alignment

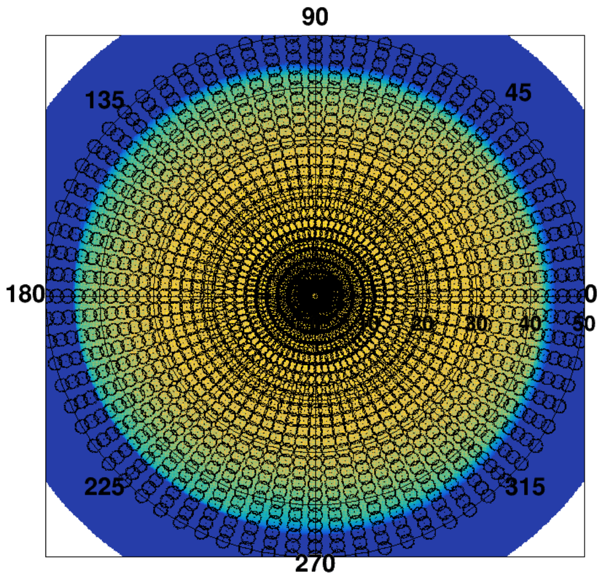
- Proper alignment is needed for comparison.
- E.g. a test pattern with crosses or blobs are used to check orientation and positioning.



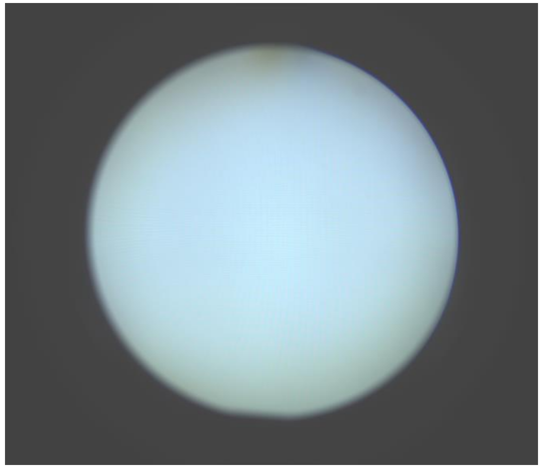


Color comparision DMS/LT

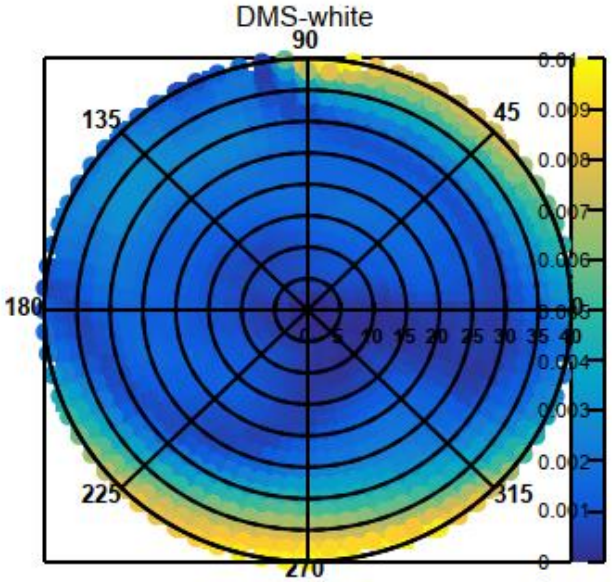
- The sample was measured with DMS and with LumiTop and the measurements are compared.
- The pixels of the LumiTop are averaged to fit the FOV of the DMS.
- DMS settings
  - Azimuth 0°-355°; Step size 5°
  - Elevation 0°-50°; Step size 2°
  - 1872 Measurement points
- LumiTop covers complete FOV **with one image (no need to move)**
- For illustration: Camera image overlayed with the measurement points of the DMS (black circles)

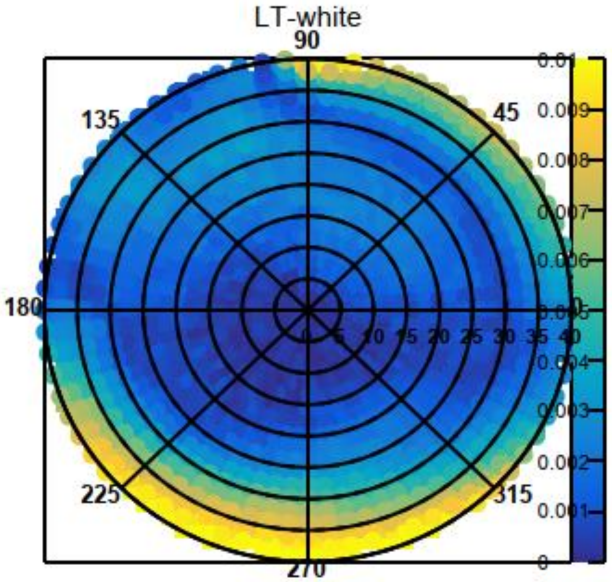


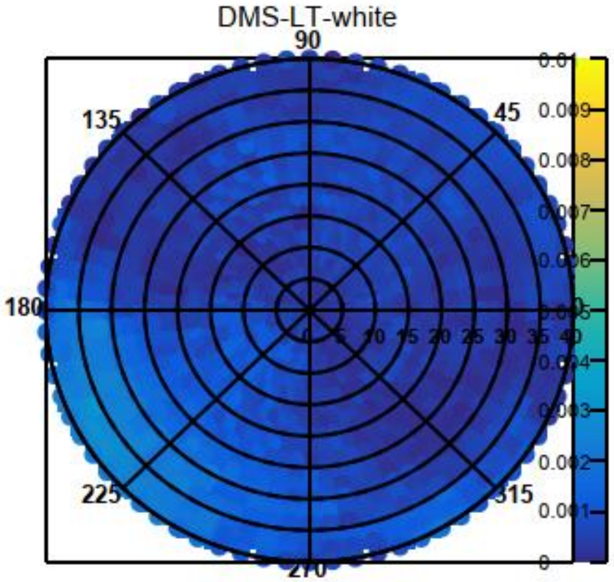
Color comparison DMS/LT - white



- DMS-white: Δ u‘v‘ relative to center
- LT-white: Δ u‘v‘ relative to center
- DMS-LT-white: Δ u‘v‘ between DMS-LT
- Good match between DMS and LT measurements



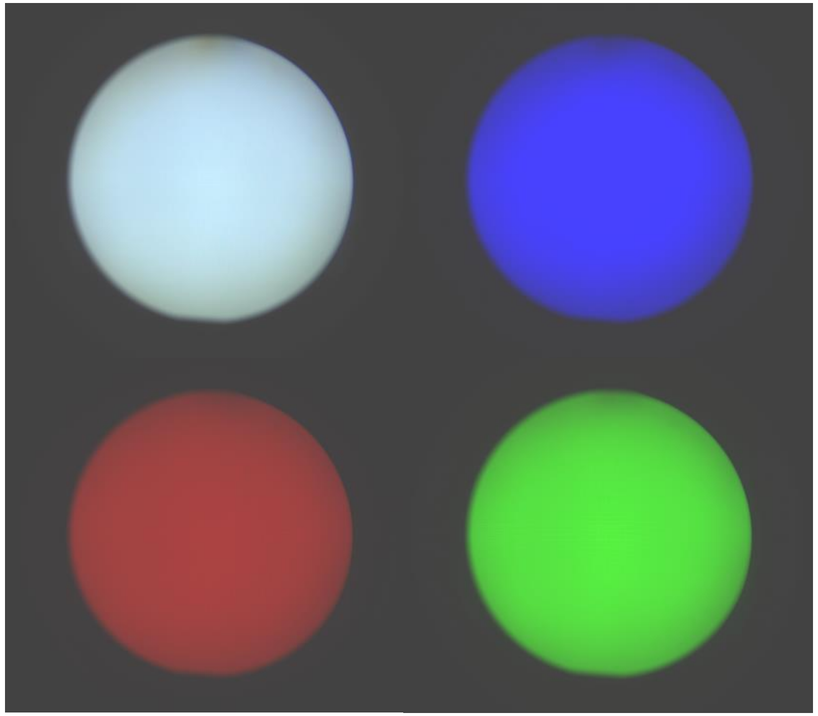


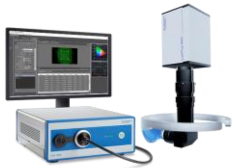


Comparison LumiTop AR/VR :: DMS ARVR

- Display measurement with Pancake optics
  - DUT NED with LCD and pancake lens

Difference between LMD and ILMD (DMS-LT) for different color pattern	Mean Δu´v´	Standard deviation Δu´v´
White	0.0005	0.0006
Red	0.0016	0.0010
Green	0.0028	0.0025
Blue	0.0016	0.0012





Conclusions

- The characterization of NED is done with an LMD and an ILMD.
  - Absolute, accurate and traceable calibration of light measurement devices enables comparability between different measurement principles.
  - No DUT specific calibration (“golden sample”) is required, even across different display technology platforms, with absolute calibration of the measurement device.
- We were able to quantitatively identify correlations between one-shot wide FOV and goniometric narrow FOV MTF measurement.
  - These correlations, based on high resolution goniometric reference measurements, could be used to speed up quality control measurements using wide FOV one-shot MTF measurements.

THAT THANK YOU

for your attention!